

**Noteпад gets an upgrade**

Noteпад now does Unix line endings and no longer shows content smooshed together in one line. <https://digit.in/jun18-A26-1>

**Costa Rica to ban fossil fuels**

Costa Rica transitions completely to renewable energy and bans fossil fuels. <https://digit.in/jun18-A26-2>

A deep dive

From modified rowboats to underwater drones, we explore the depth of the sub's journey throughout history

Arnab Mukherjee | arnab@digit.in

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ubmersible vehicles were invented before Champagne and the pendulum clock. If you consider diving

bells as submarines, early mentions go back as far as Alexander the Great! Through history, a range of applications drove the evolution of submarines. In fact, we've reached a point where they've managed to become quite autonomous and reach depths that wouldn't be possible with human passengers. We're about to explore the points of evolution that have brought submarines where they are today.

A COVER JOB

The earliest known accounts of a working-submarine goes back to 1620, crediting Dutchman Cornelius van Drebbel, a 'court inventor' for King James I of England. He modified a rowboat to be enclosed, carrying 12 oarsmen within. The boat, Drebbel, was shaped to have an angled foredeck that would make it submerge when the oarsmen propelled it forward.



Artist's depiction of the maiden public voyage of the Drebbel

Over the next 100 years, submarines built on similar principles and various kinds of propulsion methods like ballast tanks were patented and developed. This success did not escape the attention of the military around the world.

MILITARISATION

Military attention converted research about submersible vehicles into a serious pursuit. The first American military submarine was reportedly deployed in 1776. 'The Turtle' was designed by David Bushnell to accommodate a single person, and was the first to use screws for vertical and horizontal propulsion.

A similar human-powered submarine was conceptualised and built by Robert Fulton for the French Navy in 1800. Dubbed the Nautilus, it was the first known submarine to use dual-propulsion. Although it proved its capability to destroy warships using mines during demonstrations, the French government, and eventually the British as well, lost interest in the project and abandoned it. Subsequent projects, as mentioned earlier, faced aban-

donment and failures too. The next big push in submarines came with mechanised propulsion.

MECHANICAL AND ELECTRICAL POWER

Conceptually, the first autonomous submarine was the French Navy's

Plongeur. The submarine used a reciprocating engine powered by compressed air from 23 tanks at 180 psi. But the first practical and air-independent submarine, the Ictineo II, was designed by Narcis Monturiol in 1867. It ran on a combustion engine powered by peroxide and steam. The air-independent peroxide propulsion system was

designed to solve the problem of oxygen availability underwater as well as optimise other aspects.

Around the same time, the Whitehead self propelled torpedoes were developed by Robert Whitehead. Towards the end of the 19th century, the modern submarine began to take shape in the work of George Garrett, who designed and built the Resurgam in 1879 in Birkenhead, England. A combined use of wood and iron made



A cutaway depiction of David Bushnell's Turtle, 1776.